

Smart Lab Equipment & Slot Reservation Portal

WORK BREAKDOWN STRUCTURE

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SECTION: H

Project Number: 1

WBS Overview:

Smart Lab Equipment & Slot Reservation Portal

1. User Authentication & Authorization
2. Equipment Availability
3. Equipment Reservation
4. Reservation Security & Reliability
5. Equipment Return & Late Return
6. Equipment Calibration Management

Detailed Breakdown:

1. User Authentication & Authorization

- User login
- User authentication
- Role identification
- Authorization
- Access control

2. Equipment Availability

- View equipment
- Check equipment status
- View available time slots
- Display availability

3. Equipment Reservation

- Select equipment
- Select time slot
- Create reservation
- Confirm reservation
- Update reservation status

4. Reservation Security & Reliability

- Prevent double booking
- Validate simultaneous requests
- Maintain reservation consistency
- Handle unauthorized access

5. Equipment Return & Late Return

- Record equipment return
- Verify return time
- Detect late return
- Update equipment status
- Handle overdue equipment

6. Equipment Calibration Management

- Check calibration status
- Verify equipment calibration
- Update calibration information

- Prevent reservation of invalid/unavailable equipment

Requirement → Work Breakdown:

Requirement / Story	Work Breakdown
View Equipment Availability	Availability module
Reserve Equipment Slot	Reservation module
Prevent Double Booking	Reservation Security
Handle Equipment Return & Late Return	Return Management
Manage Equipment Calibration Status	Calibration Management
Reliable Concurrent Reservation Processing	Reservation Reliability
Authentication & Authorization	Security & Access Control

Jira Work Breakdown:

Epic



Story / Requirement



Task



Subtask

Epic: Equipment Availability & Reservation

Story: View Equipment Availability

Task: Develop Equipment Availability Module

Subtask: Display available equipment

Subtask: Display available time slots

Subtask: Test availability functionality

Epic: Reservation Security & Reliability

Story: Prevent Double Booking

Task: Implement reservation validation

Subtask: Check existing reservations

Subtask: Reject conflicting reservation

Subtask: Test concurrent requests

Project

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|— Kanban

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|— Epics

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|— Stories

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|— Tasks

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└─ Subtasks

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└─ Scrum

|— Sprint 1

|— Sprint 2

|— Stories

└─ Subtasks